



VIVEK
PG (I Year I Semester)
M. Tech. in Data Science
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Indian Institute of
Technology
Roorkee



Area of Interest

Machine Learning, Deep Learning, Predictive Modeling, Data Structures & Algorithms, Data Science, Agentic AI, GenAI

Education

Year	Degree/Examination	Institution/Board	CGPA/ Percentage
2026 to 2028(Expected)	M.Tech Data Science	Indian Institute of Technology, Roorkee	-
2022 to 2026	Graduate (UG) BTech Biotechnology	Sardar Vallabhbhai Patel University of Agriculture and Technology, Modipuram, Meerut, Uttar Pradesh 250110	8.250
2021	Intermediate (Class XII)	Alpine Public School, Khurja, Bulandshahr, Uttar Pradesh 203131 (CBSE Board)	92.00 %
2019	Matriculate (Class X)	D.S. Memorial Oxford Public School, Shikarpur, Uttar Pradesh 203395 (CBSE Board)	82.16 %

Internships

- Research Internship | INMAS DRDO, New Delhi February 2026 - May 2026
Description: Topic- Effect of LDI-2000 Irradiation on Gut Microbiota of C57BL/6 Mice
Analyzed 16S rRNA metagenomic data to quantify radiation-induced gut microbiome shifts and taxonomic population dynamics in murine models. Executed comprehensive wet-lab workflows encompassing bacterial phenotypic characterization, bacteriophage isolation, and kinetic assays to evaluate complex phage-host interactions.

Projects

- Microbiome Analytics Platform with RAG Assistant | Self Project June 2026
Description: Built an automated ETL pipeline (Python, Snakemake, Pandas) and an interactive Streamlit dashboard integrated with a local RAG LLM engine (LangChain, ChromaDB, Ollama) for real-time data visualization and semantic search
- AI-Powered Resume Analyzer & Scorer | Self Project July 2026
Description: Built an automated, LLM-driven recruitment tool using Python and the Groq API (openai/gpt-oss-120b) to extract structured data from job descriptions and candidate resumes. The application features a dual-prompting scoring mechanism to evaluate candidate skills, calculate a 0–100% match score, and generate ranked outputs with detailed AI verdicts, while leveraging Pydantic schemas to enforce reliable JSON extraction and eliminate LLM hallucinations.
- HelloVR.ai (Interactive AI Recruiting Assistant)
Description: Full-stack conversational recruiting assistant powered by Next.js (Vercel) and FastAPI (Render), featuring real-time streaming LLM responses, a custom terminal interface, and robust token-optimization pipelines.

Skills

Programming & Scripting: Python, Linux/WSL, AWS CLI, Git, GitHub

Data & AI Frameworks: NumPy, Pandas, Matplotlib, Seaborn, Scikit Learn, Plotly, LangChain, ChromaDB, Ollama, FastAPI

Tools & Pipelines: Streamlit, Snakemake, Groq API, Pydantic, Conda, uv

Bioinformatics & Domain: Fastp, Megahit, Kraken2, MAFFT, FastTree, WGS, 16S rRNA, Immunoassays, Cell Culture

Certifications

- Monitoring in Google Cloud Skill Badge – Google (Issued Aug 2026)
- Analyse Sentiment with Natural Language API Skill Badge – Google (Issued Aug 2026)
- Immunology – NPTEL, IIT Kharagpur (Issued Oct 2024) | Credential ID: NPTEL24BT63S655600650
- Bioreactors – NPTEL, IIT Madras (Issued Aug 2024) | Credential ID: NPTEL24BT39S141600077

